Nikita Kirnosov
Principal Engineer / Engineering Manager

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Skills

Technical Leadership, AI/ML Platforms Design, High Load Systems, Recommenders.

Languages C/C++, Python, Java, Go, SQL.

Experience

2021 Apr – Present **Staff Software Engineer / Engineering Manager**, *Google - Search*.
Created and lead a User Updates Infrastructure team powering notifications, promos, and suggestions to deliver a quarter of inorganic Search growth in 2023. Established and led cross-org collaboration of 27 engineers from Google Research, Core Infrastructure, Android, enabling 83 client team SWEs and PM work by changed user-content matching approach and adopting recent AI advances.
- Reduced new update product idea-to-MVP-metric time from a month to 2 hours and time to launch from a quarter to 2 weeks.
- Created a full cycle ML/AI ecosystem (indexing, generation, matching, training, serving).
- Augmented heuristic-based content-user matching rules with ML based content retrieval, improving user engagement by 64%.
- Introduced on-device ML to a stack, improving notification triggering quality by 27%.
- Streamlined real-time (15 minute fresh) encoder-decoder architecture recommendation models training and experimentation (impact aligned with industry expectations).
- Introduced LLMs and established Magi collaboration to power:
  - Topic updates generation: users receive meaningful updates on the followed queries (16% re-engagement increase).
  - Product-user targeting automation: market research docs are translated into platform configurations (saves several SWE-weeks per product).
- Championed the support for human voices in Search.

2018 Apr – 2021 Apr **Senior Software Engineer**, *Google - Search*.
Envisioned and lead the team implementing ML training and serving ecosystem for Search's News and Content Discovery.
- Designed and lead the implementation of unified and standardized ML infrastructure for Discover, Notifications, Suggest and user modeling teams featuring:
  - signal retrieval and rich inference serving platform (C++),
  - zero training/serving skew near real time training data generation system, reducing the feature engineering work by 70% across the org (Go, C++),
  - reinforcement learning support (in collaboration with DeepMind),
  - dynamic triggering rate optimization based on recent and long-term user activity pattern (C++).
- Scaled an existing inference serving infrastructure to serve 24x more inferences using the same resources.
- Restructured Discover Search training process to support training on TPU to achieve 40x speedup (Python).
- Launched a flexible accelerator resources acquisition and scheduling solution to improve GPU and TPU utilization by 140% (in collaboration with DeepMind).
- Created user opt-out detection system proactively stopping bad experiments (C++).

2016 Apr – 2018 Apr **Software Engineer**, *Google - YouTube*.
Built storage abstraction microservices framework with no-downtime migration support (C++) and improved YT Public Data API performance, usability, and security (Python).
- Designed and implemented storage management framework allowing the user to create a full set of standard RESTful APIs for a defined resource in several hours regardless of the storage type (C++).
- Migrated video metadata migration between Vitess and Spanner with no downtime, data loss, or client latency increase (C++).
- Analyzed data access patterns and added implicitly needed methods to the API (Python).
- Simplified fraud detection by applying Fourier transform to visualize client requests in frequency domain (Python).

2015 Sep–Oct **Fellow**, *Insight Data Science*.
Built *CityGravity.info*, a trip schedule optimization app to see the greatest number of attractions.

2010 Aug – 2016 Jan **Graduate Researcher**, *University of Arizona*, Tucson,AZ.
Developed Fortran code modeling the molecular system total wave function as an optimal combination of explicitly correlated Gaussian functions. Generalized the approach to be applied to exotic atoms and molecules and extended it to excited states. The results are published in over a dozen peer-reviewed papers and presented on multiple physics and supercomputing conferences.

2010 Jan–May **Intern**, *Joint Institute of Nuclear Research*, Dubna,Russia.
Contributed to muon detector for CMS software development by assessing signal processing code and vectorizing it.

Education

2010–2015 **Ph.D.**, *University of Arizona*, Tucson, AZ – Physics.

2005–2010 **B.S. (Hons.)**, *Saratov State University*, Saratov, Russia – Physics.